Melbourne Airbnb Market Screening

CMCE30005 Final Project Report | Group 2

Eric Huang and Maksym Xu

Project repository: https://github.com/maksym-xu/CMCE30005-TheNextChapter/tree/final-report-marking-fixes-2026-10-16

1. Introduction

Our client plans to lease homes in Greater Melbourne and let them as short stays through Airbnb. Rent and setup costs are paid before any guest income arrives, so a lease in the wrong place ties up scarce startup funds.

We recommend searching two- and three-bedroom apartments in the City of Melbourne first, with three-bedroom houses and townhouses in Yarra Ranges as a second option. These groups had the highest shares of homes with strong guest activity, and their typical homes brought in an estimated AUD 30,600–48,600 a year before costs, against AUD 24,700 for all homes studied.

Within these groups, a simple model picks homes that reach our activity target about half the time (49.6%), against one in three (32.4%) at random. It ranks only homes already on Airbnb; for other rentals, use the group order and the checks in Section 6.

2. Problem Definition and Objectives

The business question is: **where should a rent-to-rent operator with limited funds search, and which homes should it inspect first?** We set three objectives:

1. Rank property groups by the share of homes with strong guest activity, and translate this into estimated nights and revenue.
2. Test whether a simple model picks better homes within the leading groups than random picking.
3. Turn the results into a search order and a four-week inspection plan.

A **group** combines council area, dwelling type and bedroom count. A home is **busy** (it reaches our **target**) when it receives at least 30 guest reviews in the 365 days before data collection, a level reached by a quarter of separate reference homes (Section 3). Inside Airbnb (n.d.) assumes half of stays are reviewed and each lasts three nights, or the minimum stay if longer, capping estimated nights at 70% of the year (255). With three-night stays, 30 reviews means about 180 booked nights a year.

We rank on reviews rather than revenue because Inside Airbnb's revenue estimate multiplies these nights by the advertised price, and without rents a higher price does not mean a better lease. Rents, costs and profit are outside our data.

Building on the interim report's groups and target, we swap its random forest for logistic regression.

3. Data Description and Preparation

We use one Inside Airbnb snapshot of Melbourne, supplied through the subject LMS and collected 17 June–1 July 2026 (Inside Airbnb, 2026). It holds 25,728 listings with 90 fields and 1,026,690 dated reviews. We did not use the calendar, which cannot separate bookings from blocked nights.

Cleaning converted price text to numbers, treated "NA" as missing and counted amenities. We rebuilt annual review counts from review dates over exactly 365 days; the supplied count covers 366 days. Price is missing for 6,553 listings (25%) and bedrooms for 4,679 (18%); 4,477 listings (17%) have never been reviewed.

We keep entire rental units and condos (*apartments*) and entire homes and townhouses (*houses/townhouses*) with one to three bedrooms. Each home needs a nightly price of AUD 30–

1,500, which removes unpriced listings and extreme prices, and a first review at least a year before collection, so a full year of activity is visible.

So that the target is not set by the homes we then compare, we set aside about one in five hosts. Their 906 homes in the final groups set the target: about one in four received 30 or more reviews (75th percentile 29.75, rounded up; Table 1).

Table 1. From supplied listings to analysis sample

Step	Homes
All listings	25,728
Entire homes with one to three bedrooms	16,576
Apartments, condos, houses and townhouses	14,895
Priced AUD 30–1,500	11,099
First review at least a year before collection	6,891
After setting aside reference hosts	5,549
Analysis sample: 14 groups with at least 50 homes	**3,873**
Reference homes in these groups (target only)	906

The 3,873 analysis homes belong to 1,726 hosts in six council areas. The price rule mainly removes inactive listings: 77.8% of the 3,796 residential listings it removes had no review in the past year, against 17.0% of those kept.

4. Methodology

Group comparison. We compare the share of busy homes in the 14 groups. To show how much a share could move by chance, we redraw hosts at random 1,000 times and report the middle 90% of results.

Model. Logistic regression estimates the chance that a home is busy. Each input raises or lowers a score, which becomes a chance between 0% and 100%. Table 2 lists the outcome and the six inputs.

Table 2. Outcome and inputs of the model

Variable	Definition
Outcome	1 if the home had at least 30 reviews in the year, otherwise 0
Council area	One of six
Dwelling type	Apartment, or house/townhouse
Bedrooms	One, two or three
Price compared with similar homes	Nightly price relative to the group median, in steps of 10%
Accepts one-night stays	Minimum stay of one night (yes or no)
Amenity count	Amenities listed, in steps of ten

We compare this **full model** with two simpler options. The **group rate**, a group's share of busy homes, scores all its homes equally; within a group, this is **random picking**. The **reduced model** drops price and minimum stay to show what they add.

Assumptions. A host's homes tend to perform alike; allowing for this widens each effect's margin of error 1.1–2.6 times, so every test keeps a host's homes together. The inputs barely overlap (variance inflation factors below 3, against a warning level of 5). Each input is assumed to have a steady effect; with hosts grouped, checks find no reliable curve.

Testing. Each of 50 tests builds the model from 70% of hosts and scores the unseen homes of the other 30%. In each priority group, each method picks a quarter of the test homes, and we compare the share that were busy.

Two-year check. For 2,657 homes with two years of reviews, in ten groups, we chose the top three groups on the earlier year and followed them into the later year (target 34 reviews, set from the earlier year's reference homes).

5. Analysis and Results

5.1 Data exploration: most homes are quiet, and busy homes accept one-night stays

Guest activity is very uneven (Figure 1). The middle (median) home received 15 reviews, a quarter received 30 or more, and 334 homes (8.6%) received none.

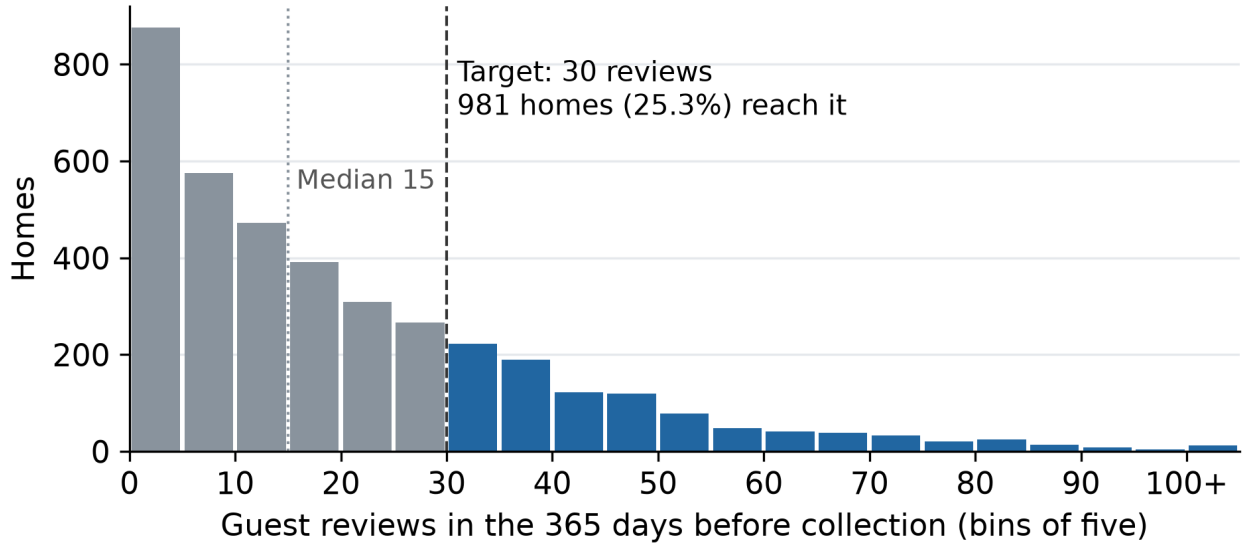


Figure 1. Annual reviews per home, 3,873 analysis homes. Source of all figures and tables: Inside Airbnb (2026); Group 2 analysis.

Table 3. Busy homes compared with the rest

Characteristic	Busy (30+ reviews)	Below 30	All homes
Homes	981	2,892	3,873
Median nightly price (AUD)	254	266	262
Median amenity count	44	43	43
Accept one-night stays	67.7%	37.5%	45.1%
Minimum stay seven nights or more	0.0%	5.7%	4.3%
In the City of Melbourne	76.9%	59.6%	64.0%
Median estimated nights booked (cap 255)	255	60	96
Median estimated revenue (AUD a year)	58,400	16,600	24,700

Busy homes were slightly cheaper and listed about as many amenities (Table 3). They were far more likely to accept one-night stays and to be in the City of Melbourne, and no home requiring seven nights or more was busy. Their estimated revenue was 3.5 times higher, but it is built from reviews.

5.2 Descriptive comparison: City apartments are the first search priority

Group rates range from 8.2% to 36.6% against 25.3% overall (Figure 2). City three-bedroom apartments lead at 36.6%, then City two-bedroom apartments (32.8%) and Yarra Ranges three-bedroom houses (30.0%; Table 4).

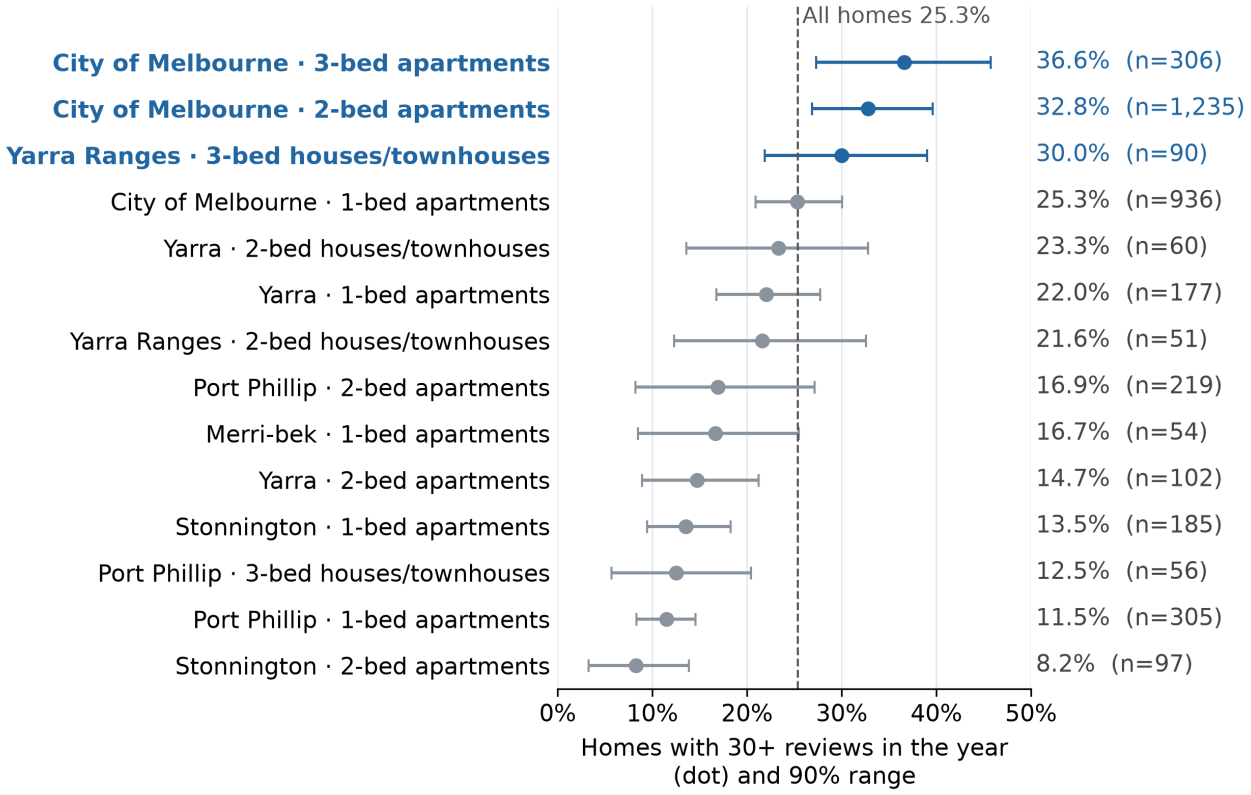


Figure 2. Share of busy homes in all 14 groups, with 90% ranges from redrawing hosts; priority groups in blue.

Table 4. Priority groups: estimated nights and revenue (medians; revenue in AUD a year)

Group	Homes	Share busy	Nights booked	Revenue	Revenue rank (of 14)	Revenue if below target
City 3-bed apartments	306	36.6%	126	48,600	1	32,100
City 2-bed apartments	1,235	32.8%	108	30,600	5	16,600
Yarra Ranges 3-bed houses	90	30.0%	120	46,100	2	31,100
All 14 groups	3,873	25.3%	96	24,700	–	16,600

The same three groups lead when the price rule or the review-history rule is removed. The two City groups are the most reliable, staying in the top three in 94% and 90% of 1,000 host redraws; Yarra Ranges houses did so in only 64%, a less certain second option.

When the two rankings differ. On revenue, Yarra Ranges houses rank second and City two-bedroom apartments fifth, mainly because the houses charge more a night (AUD 426 against 285). Since revenue rewards price and rents are unknown, we use activity to choose where to search and revenue to test each lease (Recommendation 3). Between leases that pass, choose the larger margin in each home's cash-flow plan; if margins are similar, choose the group with the higher share of busy homes.

5.3 Predictive screening: the model improves selection within the priority groups

Picking a quarter of each priority group, the full model found busy homes 49.6% of the time, against 38.6% for the reduced model and 32.4% for random picking (Figure 3). That is five busy homes in ten, not three.

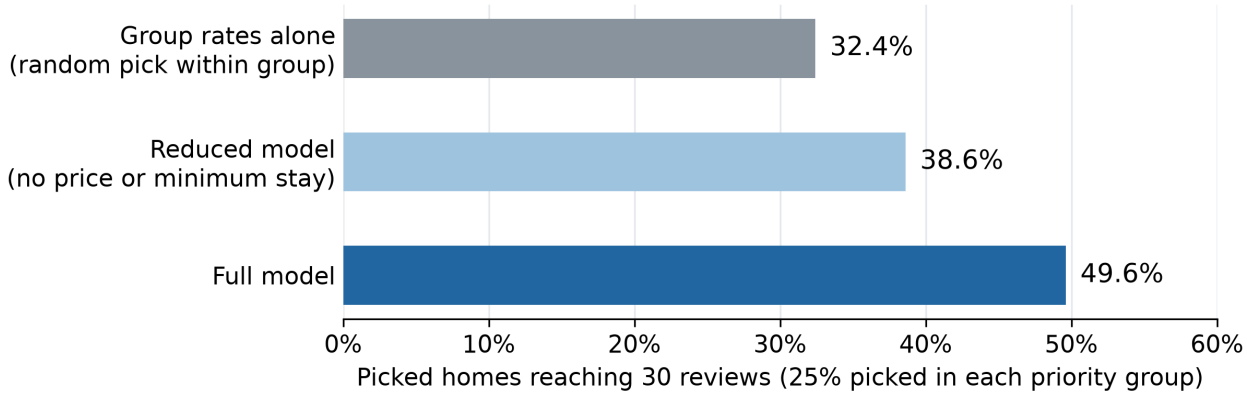


Figure 3. Share of picked homes that were busy, averaged over 50 tests.

The full model beat random picking in 46 of the 50 tests; the middle 90% of its gains ran from −0.7 to +31.8 percentage points, so results depend on the homes available.

Table 5 splits hosts into five parts and scores each home with a model built from the other four. Of picked homes, 43.7% were busy (the hit rate, or precision), against 25.3% of all homes; the picks found 43.1% of all busy homes (recall). Accuracy misleads: the model is right 71.5% of the time, yet calling every home "below target" scores 74.7% and finds no busy home.

Table 5. Hits and misses (confusion matrix): top quarter picked across all 3,873 homes

	Busy	Below target	Total
Picked by model	423	546	969
Not picked	558	2,346	2,904

With other inputs unchanged, the model fitted to all homes gives a typical City two-bedroom apartment a 42% chance of being busy if it accepts one-night stays and 18% if not. With ten more amenities the chance is 48%; priced 20% above similar homes, it is 36% (odds ratios in Appendix A). These are associations in past data, not proof that changing a setting causes more bookings.

Price and minimum stay carry most of the gain; without them, the model did worse than random in City three-bedroom apartments (33.9% against 35.9%).

5.4 Prescriptive trade-off: inspection capacity sets list length

Shorter lists have a higher hit rate; longer lists find more busy homes (Table 6).

Table 6. Inspection capacity within the priority groups (averages of 50 tests)

Homes inspected per group	Hit rate (precision)	Busy homes found (recall)	Gain over random picking (percentage points)
10%	54.5%	17.1%	+22.0
25%	49.6%	38.3%	+17.2
50%	43.1%	66.7%	+10.7

Our plan (Table 7) starts from a pool of ten candidates, whose busy share can differ widely from 49.6%.

5.5 Over time: the leading groups stayed ahead

In the earlier year, 33.6% of homes in the top three groups reached 34 reviews, against 19.9% in the other seven; this gap reflects how the groups were chosen. In the later year they still led, 27.1% against 14.9%, a gap of about 12 percentage points, or 8.7 when each host counts once within each set of groups (Figure 4).

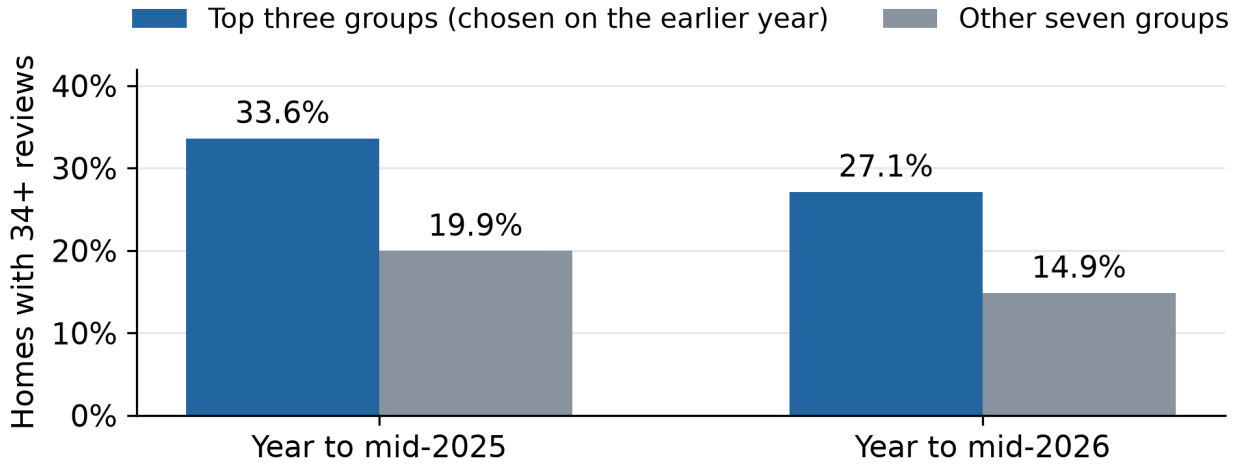


Figure 4. The same 2,657 surviving homes in two consecutive years; target 34 reviews.

6. Key Findings and Recommendations

Key findings

1. **Where:** City of Melbourne two- and three-bedroom apartments had the highest shares of busy homes and above-median estimated revenue; Yarra Ranges three-bedroom houses follow but are less certain. All three stayed ahead a year later (Figures 2 and 4, Table 4).
2. **How homes are run:** busy homes more often accept one-night stays and charge slightly less than similar homes (Table 3, Section 5.3).
3. **Which home first:** within the priority groups, the model raises the hit rate from one in three to one in two (Figure 3).

Recommendations

1. **Search City of Melbourne two- and three-bedroom apartments first**, then Yarra Ranges three-bedroom houses; between leases that pass, choose as in Section 5.2.
2. **Order candidates already on Airbnb with the model.** For an unlisted rental, price near similar listings and trial one-night stays where allowed.
3. **Drop a candidate whose weekly rent and running costs exceed its group's typical weekly estimated revenue**: about AUD 590 for City two-bedroom apartments, 930 for City three-bedroom apartments and 890 for Yarra Ranges houses (Table 4, divided by 52). Below-target homes typically earn about AUD 320, 620 and 600; keep cash for a year of that gap. Also obtain the landlord's written consent and confirm that the owners' corporation has not banned stays under 28 days (Consumer Affairs Victoria, n.d.). Such stays also attract Victoria's 7.5% short-stay levy, collected through Airbnb (State Revenue Office Victoria, n.d.).

Table 7. Four-week inspection plan

Timing and owner	Action	Decision condition
Week 1 (Operator)	Build a pool of ten candidates in the priority groups; rank listed homes within each group	Inspect top-ranked homes first, as time and budget allow; unlisted rentals follow the group order
Weeks 2–3 (Operator & Analyst)	Confirm consent and building rules; negotiate rent; obtain cost quotes	Replace homes that fail any check
Week 4 (Client)	Cash-flow plan: quotes against typical and below-target revenue (Recommendation 3)	Sign only if the plan covers setup costs and the client's required return, and the buffer covers an assumed year of below-target loss
First 30 days of operation (Operator)	Record paid nights, stay length, workload and net cash flow alongside reviews	Expand only if results match the plan

Worked example (illustrative figures). An agent offers two unlisted homes; weekly rent plus the client's quotes for cleaning, utilities, insurance and Airbnb fees total AUD 620 for a City two-bedroom apartment and AUD 700 for a Yarra Ranges three-bedroom house. The apartment's group comes first, but AUD 620 exceeds its typical AUD 590 a week, so it is dropped. The house clears its group's typical AUD 890, so it goes to rent negotiation and its own cash-flow plan, since a group median is not a forecast for one home. Earning like a below-target home (AUD 600), it would lose AUD 100 a week, about AUD 5,200 a year, which the buffer must cover. The client signs only if the plan also repays setup costs and meets its required return. New listings may start slower (Section 7).

7. Limitations and Future Work

- **Reviews are not bookings or profit.** Inside Airbnb's 50% review rate sits within published estimates of 30.5–72% (Inside Airbnb, n.d.), so the dollar figures are rough, and activity fell in the later year (Figure 4).
- **One snapshot of surviving homes.** Homes that left Airbnb are missing, and revenue uses June–July prices. New entrants, unpriced listings and outer areas are not covered.
- **Short stays inflate reviews and revenue.** A one-night minimum allows more stays for the same nights booked, so part of its effect is built into the measure; counting each stay as three nights may also overstate these homes' revenue.
- **Associations, not causes.** Settings were recorded at collection and may respond to earlier performance. We chose the one-night input after seeing no seven-night home reach the target, and chose the priority groups from all homes, so the 50 tests may overstate the gain.
- **No rent, cost or regulation data.** Recommendation 3 relies on the client's own quotes.

Future work should freeze the groups, model and selection rule, then test them on the next snapshot and the client's own homes.

8. References

Consumer Affairs Victoria. (n.d.). *Making rules to ban short stay accommodation*. Retrieved October 16, 2026, from https://www.consumer.vic.gov.au/housing/owners-corporations/rules/making-rules-to-ban-short-stay-accommodation

Huang, E., & Xu, M. (2026). *Melbourne Airbnb market screening* [Source code]. GitHub. https://github.com/maksym-xu/CMCE30005-TheNextChapter/tree/final-report-marking-fixes-2026-10-16

Inside Airbnb. (n.d.). *Data assumptions*. Retrieved October 16, 2026, from https://insideairbnb.com/data-assumptions/

Inside Airbnb. (2026). *Melbourne, Victoria, Australia: Listings and reviews* (June 2026 snapshot, via CMCE30005 LMS) [Data set]. https://insideairbnb.com/get-the-data/

R Core Team. (2026). *R: A language and environment for statistical computing* (Version 4.5.3) [Computer software]. R Foundation for Statistical Computing. https://www.R-project.org/

State Revenue Office Victoria. (n.d.). *Understanding the short stay levy*. Retrieved October 16, 2026, from https://www.sro.vic.gov.au/owning-property/short-stay-levy/understanding-short-stay-levy

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ChatGPT, Gemini and Claude Code assisted with writing and coding; Eric Huang and Maksym Xu reviewed the outputs.

Author contributions

Maksym Xu led data preparation, modelling and validation; Eric Huang led report drafting and review. Loc Le and Qihang Sun co-authored the interim report and presentation before leaving; Qihang Sun also set up the repository.

Appendix A. Model Details

We fitted the model in R (R Core Team, 2026). Group median prices and fills for missing minimum stays use training homes only. The group rate is smoothed as (group successes + 10 × overall rate) / (group homes + 10). Odds are the chance of being busy divided by the chance of not being busy. Table A1 gives each input's odds ratio from its average coefficient over the 50 training fits, and the range across fits; above 1 means a higher chance of being busy, below 1 a lower one.

Table A1. Odds ratios across 50 fits

Input (compared with)	Odds ratio	Range across fits
Accepts one-night stays (no)	3.27	2.53–4.05
Ten more amenities	1.27	1.15–1.41
Price 10% above similar homes	0.88	0.83–0.94
Council area (City of Melbourne)		
Port Phillip	0.50	0.33–0.66
Stonnington	0.48	0.26–0.68
Yarra	0.83	0.59–1.11
Yarra Ranges	1.22	0.76–1.97
Merri-bek	0.75	0.36–1.19
House or townhouse (apartment)	0.78	0.57–1.30
Two bedrooms (one)	1.16	0.89–1.41
Three bedrooms (one)	1.31	0.86–1.84

Across the 50 tests on all 14 groups, the model ranked a random busy home above a random quiet one 71% of the time, against 60% for group rates (AUC 0.71 and 0.60). Other evaluation designs are in final-report/report-2026-10-04/README.md (Huang & Xu, 2026).